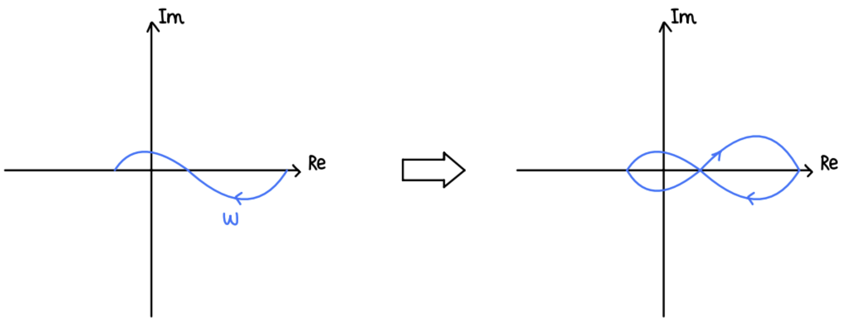
Exercise 11 - Frequency Domain Specifications and Synthesis

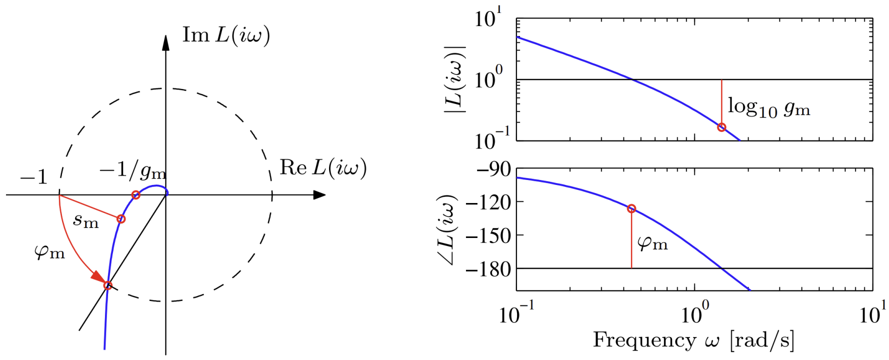
1 Introduction

Last week we saw the Nyquist stability criterion, one of the most important tools to determine the stability of a closed-loop system based on its open-loop characteristics.

Recall that the Nyquist plot is obtained by mirroring the polar plot of the frequency response about the real axis, with some special cases for poles on the imaginary axis.



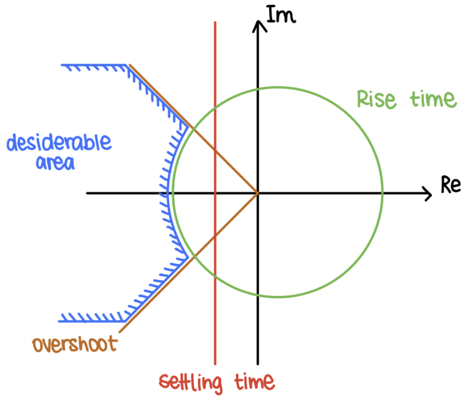
In addition, remember that the Nyquist condition can be checked from the Nyquist plot of the open-loop transfer function $L(s)$ or from its Bode plot with the help of the gain and phase margins.



Today we will focus on frequency domain specifications, which are characteristics of the closed-loop system that can be directly related to the frequency response of the open-loop system.
Furthermore, we will look at loop shaping techniques to design controllers that meet these frequency domain specifications.

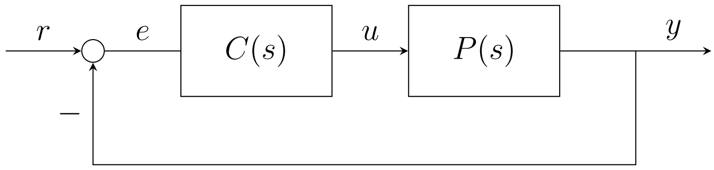
2 Frequency Domain Specifications

Frequency domain specifications are performance criteria expressed in terms of the closed-loop frequency response of the system.
They are similar to the time-domain specifications we have seen before, that resulted in feasible regions in the s-plane.

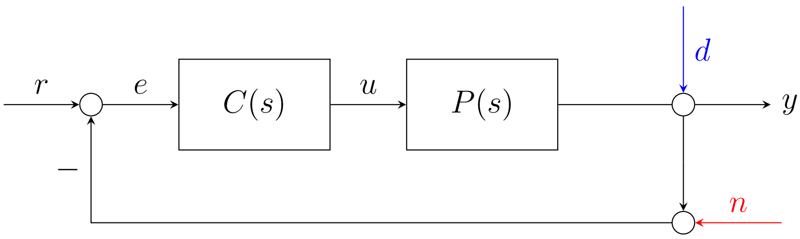


In fact, we can define frequency domain specifications that dictate/shape how the Bode plot of a system should look like in order to meet certain characteristics.

Recall the closed-loop representation using block diagrams.



Now, let's introduce some disturbance d acting on the output and some noise n due to the measurements affecting the feedback path.



Remember also the closed-loop transfer fucntion and the sensitivity transfer function that we derived some weeks ago:

$L(s) = C(s)P(s)$ Open-loop TF

$T(s) = \dfrac{L(s)}{1+L(s)}$ Closed-loop TF: (complementary sensitivity): maps $r \to y$, $n \to y$

$S(s) = \dfrac{1}{1+L(s)}$ Sensitivity: maps $r \to e$, $d \to y$

2.1 Command Tracking

Command tracking refers to the system's ability to follow a desidered input or reference signal accurately. To be more specific, for a good command tracking we want the error signal $e(t)$ to be small.

In the frequency domain, this requirement translates to having a small magnitude for sensitivity function $S(j\omega)$.

$$|S(j\omega)| \ll 1$$

As we saw before, the sensitivity function can be expressed in terms of the open-loop transfer function $L(s)$. This means that we can have this specification in terms of $L(s)$.

$$|S(j\omega)| = \left|\frac{1}{1 + L(j\omega)}\right| \ll 1 \implies |L(j\omega)| \gg \text{large value}$$

Note: This usually applies to low frequencies, since the commands we want to track are typically low-frequency signals.

2.2 Disturbance Rejection

On the other hand, disturbance rejection is the system's ability to minimize the effect of external disturbances on its output (e.g wind resistance, variance in slope). In this case, we want the effect of the disturbance $d(t)$ on the output $y(t)$ to be small.

As before, we can express s requirement in the frequency domain using the sensitivity function $S(j\omega)$ and stating that its magnitude should be small.

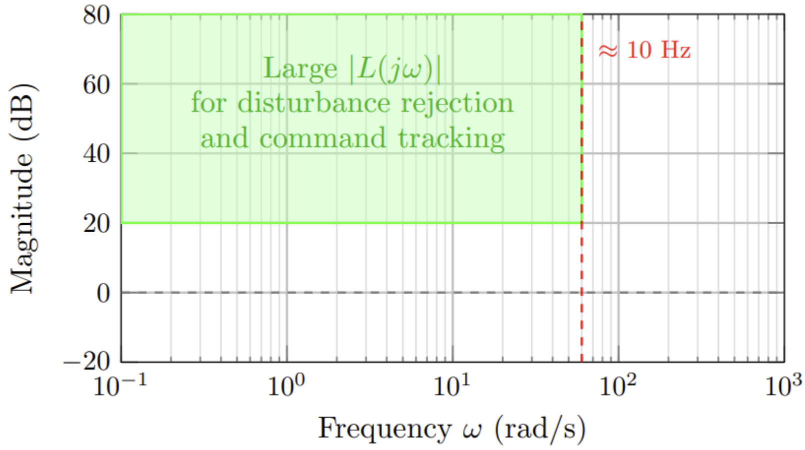
$$|S(j\omega)| \ll 1$$

Similar to command tracking, this translates to having a large magnitude for the open-loop transfer function $L(j\omega)$.

$$|L(j\omega)| \gg \text{large value}$$

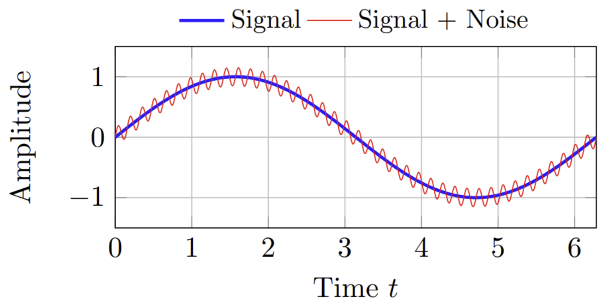
Note: Again this usually applies to low frequencies, since disturbances often affect the system at low frequencies.

All this, as for the time domain specifications, results in feasible regions in Bode plot of the open-loop transfer function $L(s)$.



2.3 Noise Rejection

Noise rejection is the system's ability to minimize the effect of measurement noise on its output. This noise could be caused for example by sensor imperfections and vibrations.



In this case, we want the effect of the noise $n(t)$ on the output $y(t)$ to be small. In the frequency domain, this translates to having a small magnitude for the complementary sensitivity function $T(j\omega)$.

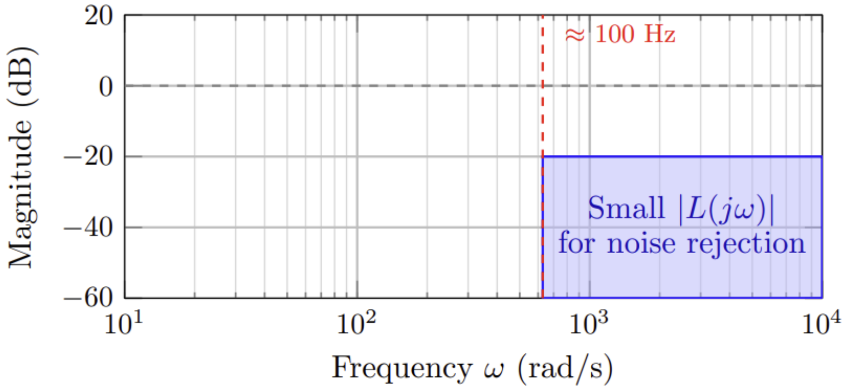
$$|T(j\omega)| \ll 1$$

As we did before, we can express this requirement in terms of the open-loop transfer function $L(s)$.

$$|T(j\omega)| = \left|\frac{L(j\omega)}{1+L(j\omega)}\right| \ll 1 \implies |L(j\omega)| \ll \text{small value} \quad \text{as} \quad |T(s)| \approx |L(s)|$$

Note: In contrast to command tracking and disturbance rejection, noise rejection usually applies to high frequencies, since measurement noise often affects the system at high frequencies.

Similarly to before, this results in feasible regions in Bode plot of the open-loop transfer function $L(s)$.



2.4 Bode obstacle course

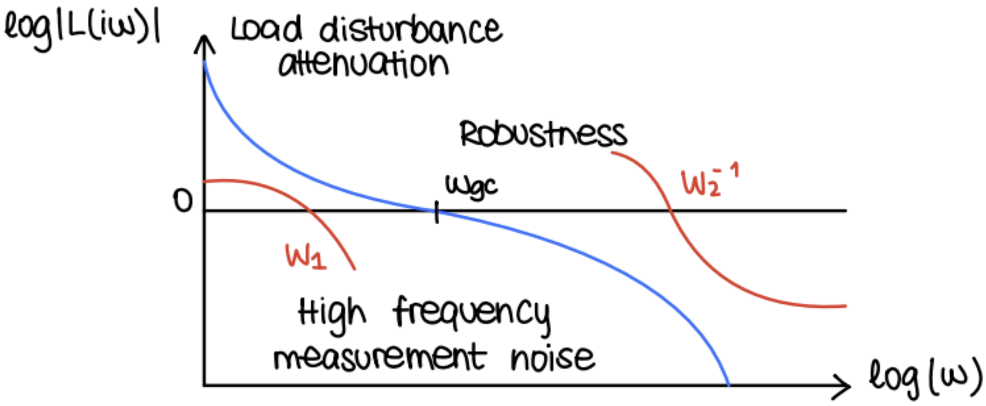
Combining all the frequency domain specifications we have seen so far, we can create a Bode obstacle course that the transfer function $L(s)$ must navigate in order to meet all the requirements.

However, we need first to introduce some limitations associated with the requirements that we have seen so far.

Ideally, we would like both command tracking/disturbance rejection and noise rejection to be small. However, this is not possible because these two cannot be achieved simultaneously. In fact, we can see that

$$S(s) + T(s) = \frac{1}{1 + L(s)} + \frac{L(s)}{1 + L(s)} = \frac{1 + L(s)}{1 + L(s)} = 1$$

This means that if we want $|S(j\omega)|$ to be small for certain frequencies, then $|T(j\omega)|$ must be large, and vice versa. Therefore, we need to find a compromise between these two requirements.
To do so, as we said before, we can visualize an "obstacle course" in the Bode plot of $L(s)$ that we need to navigate through.



We can quantify how large or small we want $L(s)$ to be, with the help of some function $W(j\omega)$. Thus, we can write:

$$|S(j\omega)| \cdot |W_1(j\omega)| < 1 \quad \Leftrightarrow \quad |1 + L(j\omega)| > |W_1(j\omega)| \approx |L(j\omega)| > |W_1(j\omega)|$$

$$|T(j\omega)| \cdot |W_2(j\omega)| < 1 \quad \Leftrightarrow \quad |L(j\omega)||W_2(j\omega)| < 1 \approx |L(j\omega)| < |W_2(j\omega)|^{-1}$$

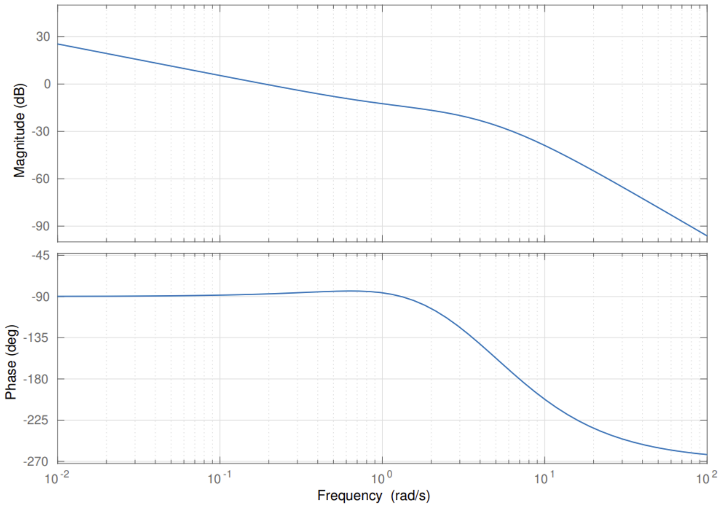
Furthemore, next to high and low-frequency behavior, we also need to take into account **bandwidth** frequency. The bandwidth tells us the maximum frequency for which the system can track commands within a factor of ≈ 0.7.
In other words, the bandwidth tells us for which maximum frequency the system can follow the reference signal with a reasonable accuracy ($|T(j\omega)| > \frac{1}{\sqrt{2}} \approx -3$ dB).

We can usually approximate the closed-loop bandwidth with the gain crossover frequency ω_{gc} of the open-loop transfer function $L(s)$.

But how can we apply all these contraints that we have seen so far to design a controller $C(s)$ that meets them? This is where loop shaping comes into play.

Exam Problem: FS23

Problem: Let L represent the open-loop transfer function of a linear time-invariant system. L is minimum-phase and open-loop stable. The Bode plot for L is given in Figure 6. The closed-loop system comprising L is shown in Figure 7 and is referred to as T.



Q23 **(0.5 Points)** What is the bandwidth BW_T of the closed-loop system T? Mark the correct answer.

☐ $BW_T \approx 0.2$ Hz.

☐ $BW_T \approx -10$ Hz.

☐ $BW_T \approx 7$ Hz.

☐ $BW_T \approx 0.03$ Hz.

Exam Problem: HS24

The control objectives are:

1. Tracking error to sinusoidal inputs of frequencies up to 1 rad/s should not exceed 1% in magnitude.
2. (Noise) Frequencies above 500 rad/s should be attenuated by at least a factor of 10.
3. The phase margin should be at least 45°.

Q37 **(0.5 Points)** Mark all correct statements.
Which of following are equivalent forms of describing control objective 1:

A $|L(j\omega)| > 100, \forall\omega > 1$ rad/s.
B $|L(j\omega)| > 100, \forall\omega < 1$ rad/s.
C $|L(j\omega)| < 0.01, \forall\omega > 1$ rad/s.
D $|L(j\omega)| < 0.01, \forall\omega < 1$ rad/s.
E $|S(j\omega)| > 100, \forall\omega > 1$ rad/s.
F $|S(j\omega)| > 100, \forall\omega < 1$ rad/s.
G $|S(j\omega)| < 0.01, \forall\omega > 1$ rad/s.
H $|S(j\omega)| < 0.01, \forall\omega < 1$ rad/s.

Q38 **(0.5 Points)** Mark all correct statements.
Which of the following are equivalent forms of describing control objective 2:

A $|L(j\omega)| > \frac{1}{10}|P(j\omega)|, \forall\omega > 500$ rad/s.
B $|L(j\omega)| > \frac{1}{10}|P(j\omega)|, \forall\omega < 500$ rad/s.
C $|L(j\omega)| < \frac{1}{10}|P(j\omega)|, \forall\omega > 500$ rad/s.
D $|L(j\omega)| < \frac{1}{10}|P(j\omega)|, \forall\omega < 500$ rad/s.
E $|T(j\omega)| > \frac{1}{10}|P(j\omega)|, \forall\omega > 500$ rad/s.
F $|T(j\omega)| > \frac{1}{10}|P(j\omega)|, \forall\omega < 500$ rad/s.
G $|T(j\omega)| < \frac{1}{10}|P(j\omega)|, \forall\omega > 500$ rad/s.
H $|T(j\omega)| < \frac{1}{10}|P(j\omega)|, \forall\omega < 500$ rad/s.

3 Loop Shaping

When designing a feedback controller, one common approach is to iteratively tune individual gains until a satisfactory closed-loop response is obtained. This so–called *trial–and–error* method consists of adjusting one parameter at a time while keeping the others fixed.
Obviously, this approach can be time-consuming and may not yield optimal results.
In this approach we assume a given plant $P(s)$ and we try to design a controller $C(s)$ such that the open-loop transfer function $L(s) = C(s)P(s)$ meets the frequency domain specifications.

$$C(s) = \frac{L_{des}(s)}{P(s)}$$

Instead, a more systematic method called *loop shaping* can be employed.
Usually, we approach this problem with some basic building blocks that can be combined to shape the open-loop transfer function $L(s)$ in order to go through the Bode obstacle course.
That way, we can design a dynamic compensator $C(s)$ that meets the desired performance criteria.

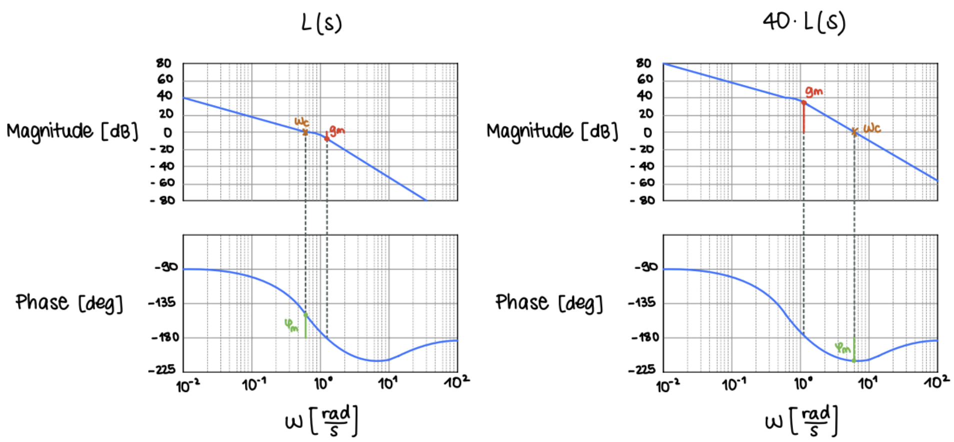
$$C(s) = C_1(s) \cdot C_2(s) \ldots C_n(s)$$

This compensators can be of different types, as for example proportional controllers, integrators, etc.

3.1 Proportional compensation

For this type of compensation, we simply multiply the open-loop transfer function $L(s)$ by a constant gain $C(s) = k$.
This results in:

- Shift up/down the magnitude, without affecting the phase.
- For stable open-loop systems, small k values yield stable closed-loop systems, while large k values can lead to instability.
- Improves tracking and disturbance rejection at low frequencies (higher magnitude at low ω) and closed-loop bandwidth (moves crossover frequency to the right).



3.2 Lead compensator

A lead compensator is a combination of a stable pole and a minimum-phase zero, where the pole is located closer to the origin than the zero. Its transfer function can be expressed as:

$$C_{lead}(s) = \frac{\frac{s}{a}+1}{\frac{s}{b}+1} = \frac{b}{a}\frac{s+a}{s+b} \qquad \text{with} \quad 0 < a < b$$

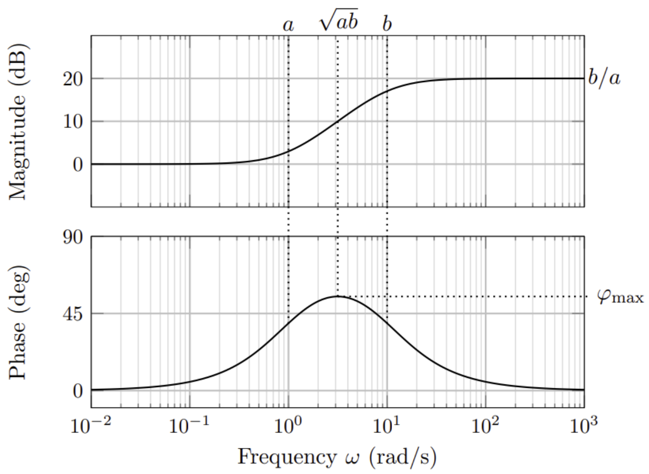
Where the parameters a and b allow us to adjust the way the lead compensator affects the magnitude and phase plot of the frequency response.
Specifically:

- Increase magintude at high frequencies by $\frac{b}{a}$, while leaving low frequencies unchanged.
- Increase slope of magnitude between a and b by +20 dB/decade.
- Increase phase around $\sqrt{ab}$ by up to 90°. The maximum phase increase is given by

$$\varphi_{max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90°$$

Graphically , this results in:

Bode Plot of a Lead Element



To design a lead compensator, we typically follow these steps:

1. Pick $\sqrt{ab}$ around the desired crossover frequency ω_{gc}.
2. Pick $\frac{b}{a}$ depending on the desired phase increase.
3. Adjust k to set the crossover frequency at the desired frequency.

In conclusion, we can say that the main use of a lead compensator is to increase the phase margin. However, it also increases magnitude at high frequencies that can worsen noise rejection.

3.3 Lag compensator

The lag compensator is is many ways the opposite of the lead compensator. It consists in the same transfer function with a stable pole and a minimum-phase zero, but in this case the zero is located closer to the origin than the pole.
Its transfer function can be expressed as:

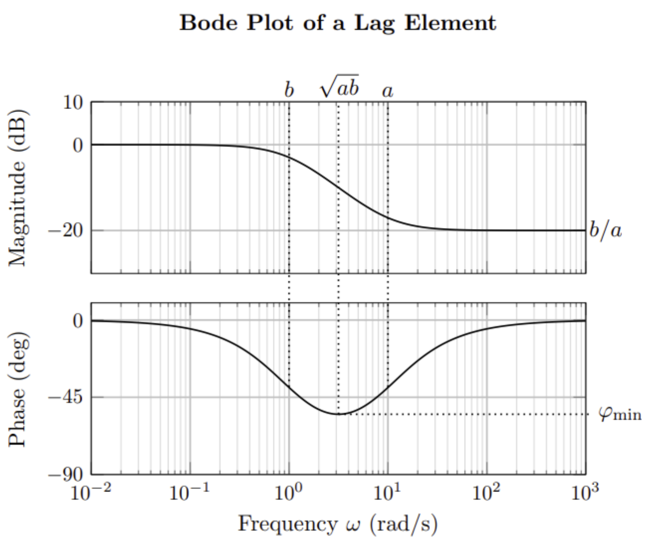
$$C_{lag}(s) = \frac{\frac{s}{a}+1}{\frac{s}{b}+1} = \frac{a}{b}\frac{s+a}{s+b} \qquad \text{with} \quad 0 < b < a$$

Here the main effects of the lag compensator on the frequency response are:

- Decrease magnitude at high frequencies by $\frac{b}{a}$, while leaving low frequencies unchanged.
- Decrease slope of magnitude between b and a by -20 dB/decade.
- Decrease phase around $\sqrt{ab}$ by up to 90°. The maximum phase decrease is given by

$$\varphi_{min} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90°$$

Graphically , this results in:



Again, to design a lag compensator, we typically follow these steps:

1. Pick $\frac{a}{b}$ depending on the desired magnitude decrease.
2. Multiply k by $\frac{a}{b}$ to leave high-frequency magnitude unchanged and increase low-frequency gain.
3. Pick a to be sufficiently small not to affect the crossover frequency and the phase margin.

We can say that the main use of a lag compensator is to improve tracking and disturbance rejection at low frequencies. However, it also decreases phase margin that can worsen closed-loop stability.

Exam Problem: HS22

Problem: Consider a lead compensator

$$C_{\text{lead}}(s) = \frac{s/a+1}{s/b+1}, \quad 0 < a < b, \tag{1}$$

used in a standard feedback architecture to control a given plant P.

Q37 **(0.5 Points)** Mark the correct answer for each statement.

Statement	True	False
A lead compensator is typically used to reduce the phase margin.		
A lead compensator can increase the sensitivity to high-frequency noise.		
The slope of the magnitude at frequencies between a and b is approximately +20 dB/decade.		

Q38 **(1.5 Points)** A lead compensator as given in Equation (1) is to be used in order to increase the phase of the open-loop system $P(s) \cdot C_{\text{lead}}(s)$ by at most 30° at a frequency of 10 rad/s.

Which choice of the parameters a and b satisfies the aforementioned requirement? Mark the correct answer.

- ☐ $a = \frac{10}{\sqrt{3}}$, $b = 10\sqrt{3}$
- ☐ $a = 1\sqrt{3}$, $b = 100\sqrt{3}$
- ☐ $a = 10\sqrt{3}$, $b = \frac{10}{\sqrt{3}}$
- ☐ $a = 100\sqrt{3}$, $b = 1\sqrt{3}$

3.4 Why PID controllers work so well?

Some weeks ago we said that PID controllers are widely used in control systems because they work well in practice. But why is that?
To understand the reason behind this, let's recall the transfer function of a PID controller:

$$C_{PID}(s) = k_p + \frac{k_I}{s} + k_D s$$

Where k_p is the proportional gain, k_I is the integral gain and k_D is the derivative gain. We can say that a PID controller is actually a coombination of a lag and a lead compensator, plus a proportional gain.
In fact, if we consider the following implementable PID controller

$$C_{PID}(s) = K \frac{\left(\frac{s}{z_1}+1\right)\left(\frac{s}{z_2}+1\right)}{s\left(\frac{s}{p}+1\right)}$$

We can see that:

- The term $\frac{\left(\frac{s}{z_1}+1\right)}{s}$ acts as a lag compensator with a pole at $s = 0$
- The term $\frac{\left(\frac{s}{z_2}+1\right)}{\left(\frac{s}{p}+1\right)}$ acts as a lead compensator with a pole at $s = -p$

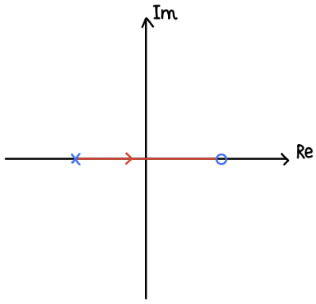
3.5 Limitations

The tools we have seen today work well for stable, minimum-phase systems. However, they have some limitations when dealing with unstable or non-minimum phase systems.

In fact, for these types of systems, we need to be more careful when designing the controller, as the presence of right-half plane poles or zeros can significantly affect the system's behavior.

Loop shaping for non-minimum phase systems

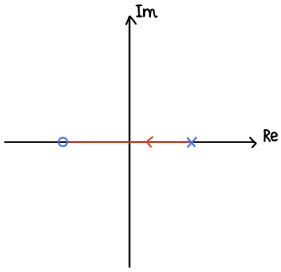
Recall that, as we saw from the root locus, closed-loop poles approach the open-loop zeros as the gain increases. Therefore, if we have a non-minimum phase zero (RHP zero), the closed-loop poles will also approach it and the closed-loop system might become unstable.



Thus for large gains, the closed-loop system might become unstable. Furthermore, a RHP zero also limits the maximum crossover frequency we can achieve, as it reduces the phase margin. This will lead to a slow closed-loop response.

Loop shaping for open-loop unstable systems

If we look at the root locus of an open-loop unstable system, we can see that as the gain increases, the pole will move to the left-half plane, making the closed-loop system stable.



This means that we need to have a high gain to stabilize the system. This, in real life, means that we have fast and strong actuators. Furthermore, in this case, the crossover frequency becomes large.

With this said, we can say that this should give us a sense of how, for many systems, there are clear performance limitations.
In other words certain requirmentes can never be met.